

1.3 Subgroup, Quotient Group

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Definition 0.1. Let G be a topological group and $H \subseteq G$. Then H is a subgroup of topological group G if H is a subgroup of group G and H is a closed set in topological space G .

Remark 0.1. H is a topological group equipped with induced topology on G . A subgroup of an abstract group which is a topological group is a topological group. This works as restriction of continuous maps is continuous.

Definition 0.2. A normal subgroup N of topological group G if N is a normal subgroup of abstract group G .

Proposition 0.1. Let G be a topological group. Let H be a subgroup of abstract group G .

- (i) $\overline{H}$ is a subgroup of top. group G .
- (ii) If H is normal subgroup of G then $\overline{H}$ is normal.

Proof. (i) Observe $\overline{H} = \bigcap_{V \in \mathcal{V}} HV$ where $\mathcal{V}$ is collection of all nbds of e . As $H \subseteq \overline{H}$ hence $\overline{H} \neq \emptyset$. For $x, y \in \overline{H}$ we want to show $xy^{-1} \in \overline{H}$. We know $m : G \times G \rightarrow G$ is continuous, where $m(x, y) = xy^{-1}$. Observe $m^{-1}(\overline{H})$ is closed, and $H \times H \subseteq m^{-1}(\overline{H})$. Hence $\overline{H} \times \overline{H} = \overline{H \times H} \subseteq \overline{m^{-1}(\overline{H})}$ and this implies $m(\overline{H} \times \overline{H}) \subseteq \overline{H}$. $\square$

We construct a mapping

$$n : G \times G \rightarrow G, \quad (x, y) \mapsto xyx^{-1}$$

and observe that n is a continuous map. This implies that $n^{-1}(\overline{H})$ is closed and similar to (Ci).

$$H \times H \subseteq n^{-1}(\overline{H}) \implies \overline{H} \times \overline{H} = \overline{H \times H}$$

is a subset of $n^{-1}(\overline{H})$. Hence

$$n(\overline{H} \times \overline{H}) \subseteq \overline{H}.$$

Definition 0.3. * Set of all left cosets set G/H . We have a natural map or projection

$$\pi : G \rightarrow G/H, \quad x \mapsto xH.$$

Remark 0.2. If G is a topological group, we will topologize G/H assuming H is closed. Construct

$$\mathcal{T}_{G/H} = \{O \subseteq G/H \mid \pi^{-1}(O) \in \mathcal{T}_G\},$$

i.e., O is open if and only if $\pi^{-1}(O)$ is open in G .

which ensures π is a continuous map, moreover a quotient map where π is surjective by construction.

Lemma 0.1. G/H is a T_1 -space, and π is open.

Proof. Since H is closed, then αH is closed by homeomorphism. Observe $G - \alpha H$ is open in G . Let $\tilde{\alpha} = \alpha H \in G/H$. Observe $\pi^{-1}(G/H - \tilde{\alpha}) = G - \alpha H$. Hence $\tilde{\alpha}$ is closed, G/H is T_1 .

We want to show π is an open map. Let O be an open map in G/H then $\pi(O)$ is open in G/H iff $\pi^{-1}(\pi(O))$ is open in G . Now $OH = \pi^{-1}(\pi(O))$ and $OH = \bigcup_{\alpha \in O} \alpha H$. Hence OH is open. $\square$

$\pi^{-1}(\pi(O))$ is open which implies $\pi(O)$ is open.

Lemma 0.2. G/H is a T_2 space.

Proof. Consider $xH \neq yH$ in G/H . Now $x \in xH$ and $x \notin G \setminus yH$ which is open as H is closed and yH is closed. Now, let V be a neighborhood of x such that $x \in V \subset G \setminus yH$. Observe $Vx \cap yH = \emptyset$ implies $VxH \cap yH = \emptyset$ otherwise $vh_1 = gh_2$ and $v = gh_2h_1^{-1}$ implying $Vx \cap yH \neq \emptyset$. Now, we can find a neighborhood V_1 of V such that $V_1^{-1}V_1 \subset V$ therefore $V_1^{-1}V_1xH \cap yH = \emptyset$. This implies $V_1xH \cap V_1yH = \emptyset$, which is $\pi(V_1x) \cap \pi(V_1y) = \emptyset$ and by previous lemma $\pi(V_1x)$ and $\pi(V_1y)$ are open sets and $xH \in \pi(V_1x)$ and $yH \in \pi(V_1y)$. $\square$

Remark 0.3. We call G/H the quotient space where H is a closed subgroup of H , has equipped with quotient topology.

Lemma 0.3. If G is a topological group, H is normal subgroup, the quotient topology equipped on the quotient group G/H is a topological group.

Proof. We want to show $\varphi_1 : G/H \times G/H \rightarrow G/H$ defined by

$$\varphi_1(\alpha_1, y_1) \mapsto \alpha_1 y_1^{-1}$$

is continuous.

And $\varphi : G \times G \rightarrow G$ defined by

$$\varphi(\alpha, y) \mapsto \alpha y^{-1}$$

is continuous. $\square$

$$\begin{array}{ccc} G \times G & \xrightarrow{\pi \times \pi} & G/H \times G/H \\ \downarrow \pi & & \downarrow \psi \\ G & \xrightarrow{\varphi} & G/H \end{array}$$

Observe that $\psi_1(\pi \times \pi) = \psi\pi$ and π, ψ are continuous. Hence $\psi_1(\pi \times \pi)$ are continuous.

Let O_1 be an open subset of G/H then $\psi_1(\pi \times \pi)^{-1}(O_1)$ is open. As $\pi \times \pi$ is an open map, observe that $\pi \times \pi(\pi \times \pi)^{-1}\psi_1^{-1}(O_1)$ is open. Hence $\psi_1^{-1}(O_1)$ is open.

Proof. $\square$

Lemma 0.4. Let G be a topological group, and H be a subgroup.

- (a) If G is compact then H and G/H are both compact.
- (b) If G is locally compact then H and G/H have both locally compact.

Proof. (a) H is a closed subset of a compact set and $G/H = \pi(G)$ where G is compact and π is a continuous mapping. Hence a continuous map of a compact set is compact.

(b) A closed subset of a locally compact set is locally compact.

Let S be locally compact, $T = \overline{S}$ and $T \subseteq S$. For $p \in T \cap S$, there exists a neighborhood of p in S , $\overline{U_p}$ such that $\overline{U_p}$ is compact. Observe for $p \in T$, there exists an open set $T \cap U_p$ in T such that $\overline{T \cap U_p} = \overline{T} \cap \overline{U_p}$ is compact as $\overline{T \cap U_p}$ is a closed subset of $\overline{U_p}$ which is compact. Hence H is locally compact.

Goal: G/H is locally compact.

□

Let $q_i \in G \setminus H$ and let U_i be a neighborhood of q_i . Let

$$U = \pi^{-1}(U_i),$$

and let $x \in G$ such that $\pi(x) = q_i$. As G is locally compact, we get a neighborhood O of x such that $\overline{O}$ is compact and $\overline{O} \subseteq U$ by the proposition given below. Then we have $\pi(\overline{O}) \subseteq \pi(U) = U_i$. If $O_i = \pi(O)$ is a neighborhood of q_i . And $\pi(\overline{O})$ is a continuous image of a compact set, hence we get $\pi(\overline{O})$ is compact. Since G/H is T_2 -space $\pi(\overline{O})$ is closed which implies

$$O \subseteq \overline{O} \Rightarrow O_i \subseteq \pi(\overline{O}) \quad \text{and} \quad \overline{O_i} \subseteq \pi(\overline{O}) = \pi(\overline{O}) \quad \text{which is compact, so is } \overline{O_i}.$$

Thus O_i is a neighborhood of q_i such that $\overline{O_i}$ is compact. It follows G/H is locally compact.

Corollary 0.1. If G is locally compact, then G/H is normal or T_4 .

Proposition 0.2. For any topological group G , let U be a neighborhood of e , there exists a neighborhood of e V such that $V \subseteq U$.

Proof. By properties (5) and (6) we know that for every neighborhood of e , U , we have a symmetric neighborhood of e V such that $V^2 \subseteq U$. Now $x \in \overline{V} = \bigcap_{w \in V} wV$ where V^2 is the set of all neighborhoods of e , this implies $x \in \overline{V} \subseteq V^2 \subseteq U$. □